[1] ① Notice $y\frac{1}{X}\cdot 1_{\{X>y\}} \le 1_{\{X>y\}} \le 1$, by dominated convergence theorem, we have

$$\lim_{y\to\infty} yE\left[\frac{1}{X}1_{\{X>y\}}\right] \le \lim_{y\to\infty} E\left[1_{\{X>y\}}\right] = E\left[\lim_{y\to\infty} 1_{\{X>y\}}\right] = E[0] = 0$$

② Since. $\lim_{y\to 0} y\cdot\frac{1}{X}\cdot 1_{\{X>y\}} = 0$, by dominated convergence theorem, we have.

$$\lim_{y\to 0} yE\left[\frac{1}{X}1_{\{X>y\}}\right] = E\left[\lim_{y\to 0} y\cdot\frac{1}{X}\cdot 1_{\{X>y\}}\right] = E[0] = 0$$

[2] If $\|X\|_p$ or $\|Y\|_q = 0$, then $|XY| = 0$, a.e. So we only need to prove the statement when $\|X\|_p$ and $\|Y\|_q > 0$ and by further dividing both sides by $\|X\|_p\|Y\|_q$, when $\|X\|_p = \|Y\|_q = 1$. So we assume now that. $\|X\|_p = \|Y\|_q = 1$

Fix $y>0$, we consider following function $\varphi(x)$:

$$\varphi(x) = x^p/p + y^q/q - xy \quad \text{for } x>0.$$

We know $\varphi'(x) = x^{p-1} - y$ and $\varphi''(x) = (p-1)x^{p-2}$, so $\varphi(x)$ has a minimum at $x_0 = y^{1/p-1}$. $q = p/(p-1)$ and $x_0^p = y^{p/p-1} = y^q$.

So $\varphi(x_0) = y^q(1/p + 1/q) - y^{1/p-1} = 0$.

Since x_0 is the minimum, it follows that $xy \le x^p/p + y^q/q$ (*)

for $x, y>0$. Obviously (*) is also true for $x, y \ge 0$. Let $x=|X|$, $y=|Y|$; then we have

$$E[|XY|] \le \frac{1}{p} + \frac{1}{q} = 1 = \|X\|_p\|Y\|_q.$$

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[3] $$\sum_{n=1}^{\infty} P(A_n) = \sum_{n=1}^{\infty} n^{-\alpha} = \begin{cases} +\infty & \alpha \le 1 \\ < \infty & \alpha > 1 \end{cases}$$

By Lemma 3.1.8, we have $P(\limsup_{n\to+\infty} A_n) = \begin{cases} 0 & \text{if } \alpha > 1 \\ 1 & \text{if } \alpha \le 1 \end{cases}$

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[4] Suppose $\sum_{n=1}^{\infty} P(A_n) < \infty$, so we know

$$\left| \sum_{n=1}^{\infty} \ln(1 - P(A_n)) \right| < \infty ,$$

by the result $\ln(1+x) \sim x$ when $x \to 0$.

On the other side, A_n independent $\Rightarrow$ A_n^c independent

So $$-\sum_{n=1}^{\infty} \ln(1 - P(A_n)) = -\sum_{n=1}^{\infty} \ln(P(A_n^c)) = -\lim_{N\to\infty} \sum_{n=1}^{N} \ln(P(A_n^c))$$

$$= -\lim_{N\to\infty} \ln\left(\prod_{n=1}^{N} P(A_n^c)\right) = -\lim_{N\to\infty} \ln\left(P\left(\bigcap_{n=1}^{N} A_n^c\right)\right)$$

$$= +\lim_{N\to\infty} \ln\left(1 - P\left(\bigcup_{n=1}^{N} A_n\right)\right) = -\infty ,$$

Which is a contradiction to the assumption.

Therefore $\sum_{n=1}^{\infty} P(A_n) = \infty$ and then $P(\limsup_{n\to\infty} A_n) = 1$ by Theorem 3.1.8.

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